Predicting Diabetes Risk Using the CDC-NHANES Dataset: A Machine Learning Approach

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***Abstract*—Diabetes mellitus is a major global health problem affecting millions of people and leading to serious complications if not detected early. This study uses the CDC-NHANES dataset to create machine learning models capable of classifying individuals as 'Normal,' 'Prediabetes,' or 'Diabetes' based on comprehensive health indicators. We trained and evaluated four models: logistic regression, support vector machines (SVM), random forest, and artificial neural networks (ANN). The Random Forest model had the highest accuracy of 86.84%, highlighting the effectiveness of ensemble methods in diabetes risk prediction. The study emphasizes the value of data-driven approaches for improving early detection and informing targeted healthcare interventions.**

***Index Terms*—Machine Learning, Diabetes Prediction, CDC-NHANES, Random Forest, Support Vector Machines, Artificial Neural Networks, Public Health**

I. Introduction

Diabetes mellitus is a chronic condition that causes damage to the heart, blood vessels, eyes, kidneys, and nerves [1]. According to a new study published in The Lancet, more than 800 million adults worldwide had diabetes in 2022, and significantly, 59 percent of them aged 30 years or older did not receive treatment [2]. Early detection and management are critical for avoiding serious complications and reducing healthcare costs.

Despite advances in medical diagnostics, a significant proportion of individuals remain undiagnosed or unaware of their prediabetic status due to asymptomatic early-stage diabetes [3]. Machine learning presents promising opportunities for improving early detection through predictive modeling using readily available health data.

The CDC-NHANES (National Health and Nutrition Examination Survey) dataset collects health-related data from a representative sample of the US population [4]. Using this dataset, the aim is to develop predictive models for diabetes risk assessment, which could aid in timely interventions and resource allocation.

A. Research Questions and Objectives

This study addresses the following research questions:

1) **RQ1**: Can machine learning models accurately classify individuals into 'Normal,' 'Prediabetes,' or 'Diabetes' categories using the CDC-NHANES dataset?
2) **RQ2**: Which health indicators are the most significant predictors of diabetes risk in the CDC-NHANES dataset?
3) **RQ3**: How does the performance of different machine learning algorithms (Logistic Regression, SVM, Random Forest, ANN) compare in predicting diabetes risk?
4) **RQ4**: What are the practical implications of using these predictive models for early detection and public health strategies?

Objectives and Potential Benefits: The primary goal of this study is to create robust machine-learning models that categorize people into different diabetes risk categories (Normal, Prediabetes, and Diabetes) based on their health and demographic information. By identifying key risk factors, we hope to improve early detection capabilities and support public health initiatives. The potential benefits include improved patient outcomes through timely interventions, optimal resource allocation by focusing on high-risk groups, informed policymaking, and the promotion of personalized healthcare through tailored advice based on individual risk profiles.

II. Related Work

The use of machine learning (ML) techniques to predict diabetes has been extensively studied, with diverse datasets and methodologies used to improve predictive accuracy and early detection. A lot of attention has been given to the National Health and Nutrition Examination Survey (NHANES) dataset, which is valued for its comprehensive and representative health data [4].

Dinh et al. conducted a thorough analysis of NHANES data to develop machine learning models for detecting cardiovascular and diabetes conditions. Their research compared various algorithms, including logistic regression, support vector machines, random forest, and gradient boosting, before creating a weighted ensemble model to improve detection accuracy. Diabetes predictors identified include waist circumference, age, self-reported weight, leg length, and blood pressure readings [5], highlighting the significance of demographic and

physiological factors in disease prediction. This paper guided the analysis in determining which key features to include in the models to improve prediction accuracy, such as blood pressure readings, age, and waist circumference.

Furthermore, the selection of relevant features was informed by various studies, including the inclusion of the variable "Family History of Diabetes". An analysis of NHANES data from 1999 to 2004 found a strong graded relationship between familial risk and diabetes prevalence. The study discovered that people with a moderate familial risk were 2.3 times more likely to develop diabetes. In contrast, those with a high familial risk were 5.5 times more likely than people with an average familial risk. These associations remained significant even after adjusting for factors such as gender, race/ethnicity, age, BMI, hypertension, income, and education [6].

In terms of dietary factors, researchers used NHANES data to investigate the relationship between total sugar intake and diabetes risk. Sneed et al. [7] investigated whether total added sugar consumption was associated with an increased risk of prediabetes among US adults. The results showed that total added sugar intake was not significantly associated with prediabetes risk, implying that other dietary components or lifestyle factors may play a larger role in prediabetes development. Nonetheless, in the models, total sugar intake was included alongside other dietary variables to control for its potential effects.

In their study [8], Ordoñez-Guillen et al. used clustering methods to identify subtypes of Type 2 Diabetes Mellitus, leading to a better understanding of the disease's complexity and distinct patient profiles. This highlights the importance of using clustering and association rules mining techniques to understand patient profiles and the characteristics that affect different stages of the disease.

In a comparative study, Kumar et al. [9] utilized various ML algorithms such as Neural Networks, Linear Discriminant Analysis, Fine K-Nearest Neighbors, Fine Tree, Boosted Tree, and Bagged Tree to predict the development of diabetes at an early stage. The Bagged Trees algorithm outperformed the others, achieving the highest accuracy of 97.7%, demonstrating the effectiveness of ensemble methods in improving predictive performance. While in this study Random Forest was used, also achieving the highest overall test performance.

Another study proposed an ensemble approach based on machine learning to predict early-stage diabetes risk, addressing challenges with disease prediction and a lack of awareness about risk factors before diabetes diagnosis. The study used classification and ensemble approaches with varying degrees of accuracy, including Random Forest, Gradient Boosting, Decision Tree, K-Nearest Neighbor, Logistic Regression, and Support Vector Machine, highlighting the importance of model selection in predictive analytics [10].

Recent studies have also addressed the challenge of class imbalance in medical datasets. Sadeghi et al. [11] investigated diabetes risk prediction in the presence of class imbalance, highlighting the need for strategies such as SMOTE to improve model performance on minority classes.

In summary, combining machine learning techniques with large datasets such as NHANES has significantly improved diabetes predictive modeling. The related work influenced the selection of necessary features and appropriate models, emphasizing the importance of addressing class imbalance and using ensemble methods to improve predictive performance.

III. Proposed Methodology and Machine Learning Techniques

A comprehensive machine learning pipeline for predicting diabetes risk is proposed. The pipeline consists of data preprocessing, feature engineering, handling class imbalance, model training, and evaluation. Figure 1 shows the overall workflow.



Fig. 1. Machine Learning Pipeline Flow Chart

A. Data Preprocessing

Data preprocessing involves merging the relevant NHANES data files, handling missing values using Multiple Imputation by Chained Equations (MICE), and encoding categorical variables. Features are standardized using the *StandardScaler* to ensure all variables contribute equally to the models.

B. Handling Class Imbalance

To address the class imbalance inherent in the dataset, the Synthetic Minority Oversampling Technique (SMOTE) [12] was applied. SMOTE generates synthetic examples for minority classes ('Prediabetes' and 'Diabetes') to balance the class distribution, which is crucial for training models that are not biased toward the majority class.

C. Machine Learning Algorithms

Four machine-learning algorithms were used:

- **Logistic Regression (LR):** A linear statistical model that predicts the probability of each class using a logistic function. It is extended to multiclass classification via the one-vs-rest approach. LR is a suitable baseline model due to its simplicity, interpretability, and robustness to small datasets [12].

- **Support Vector Machines (SVM):** SVM separates classes by identifying an optimal hyperplane in feature space. For this dataset, SVM uses a radial basis function (RBF) kernel to capture non-linear relationships. This method was chosen because it is effective in high-dimensional spaces due to its ability to create flexible decision boundaries. However, when dealing with imbalanced classes, standard SVMs may exhibit bias towards the majority class, leading to suboptimal performance [12].

- **Random Forest (RF):** An ensemble learning technique that constructs multiple decision trees and aggregates their outputs. RF is robust to overfitting, handles high-dimensional datasets effectively, and provides feature importance rankings, enabling interpretability of the model's decisions [12].

- **Artificial Neural Networks (ANN):** A computational model inspired by the human brain's neural networks. ANN captures complex nonlinear relationships by utilizing multiple layers of neurons. This technique is ideal for identifying intricate patterns in the dataset [12].

D. Model Training and Evaluation

The dataset was splitted into training (60%), validation (20%), and test (20%) sets. Hyperparameters are optimized using Bayesian Optimization [13] combined with cross-validation to improve model performance and generalizability.

E. Example Trace Through the Pipeline

Consider a sample data point that contains missing values for 'Family History of Diabetes' and 'Glycohemoglobin'. During preprocessing, MICE imputes these missing values using other correlated features. The data point is then standardized, and if it falls into the minority class, SMOTE may generate synthetic examples that are similar to it. The preprocessed data point is fed into each model, which assigns it to one of the risk categories ('Normal,' 'Prediabetes,' or 'Diabetes') according to the model's classification rules.

IV. Experimental Evaluation

A. Dataset Description

The CDC-NHANES dataset utilized in this study comprises five files: Demographic, Laboratory, Examination, Dietary, and Questionnaire data. After merging based on the unique identifier (SEQN), the dataset contains 9813 variables with 26 features relevant to diabetes risk assessment before preprocessing. Table I describes key variables selected for classification.

It is also important to note that minors were excluded from the dataset because they provided the majority of the missing values. Specifically, the Family history diabetes feature was completely missing, affecting all of the results, as shown in Figure 2.

B. Experimental Settings

First, these are defined as:

- **Independent Variables:** Demographic, nutritional, and clinical variables, such as BMI, blood pressure, glycohemoglobin, and caloric intake.
- **Dependent Variable:** Diabetes status, derived from a combination of laboratory-confirmed diabetes, prediabetic markers, and self-reported health conditions.

Then the following hypotheses are tested:

- **H1**: Applying SMOTE will improve the classification performance for the 'Prediabetes' class without significantly reducing overall accuracy.
- **H2**: Ensemble methods like Random Forest will outperform individual classifiers in predicting diabetes risk.

Since it is mainly a classification problem with an unbalanced dataset the following are the metrics used to measure the performances of the models:

- **Accuracy**: This is the ratio of the number of correct predictions to the total number of predictions.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- **Precision, Recall, and F1-score**: Precision is the ratio of true positives to the sum of true positives and false positives. Recall (or sensitivity) is the ratio of true positives to the sum of true positives and false negatives. The F1-score is the harmonic mean of precision and recall. These metrics are useful when the classes are imbalanced, like in this case.

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

- **Confusion Matrix**: Visualizes the model's performance in terms of true positives, true negatives, false positives, and false negatives, providing detailed insights into classification errors.

- **ROC-AUC:** The area under the receiver operating characteristic curve, evaluating the trade-off between sensitivity and specificity across thresholds.

So the best model was chosen taking into consideration a combination of these.

C. Results

*1) **Clustering**:* Clustering techniques were used to identify natural groupings in the data. K-Means clustering identified three somewhat distinct clusters based on health indicators: low-risk, medium-risk, and high-risk. The Silhouette Score and Elbow Method were used to assess cluster consistency and determine the ideal number of clusters. Agglomerative Clustering confirmed the findings of K-Means. Figure 3 visualizes the clustering results using Principal Component Analysis (PCA) in three dimensions. Even though the clusters were not perfectly separated, it was decided to include them to assist the model in classifying the outcome variable.

*2) **Association Rule Mining**:* The Apriori algorithm was employed to discover frequent item sets and generate association rules among categorical variables. Continuous variables were discretized into categorical bins. Significant associations identified include a strong correlation between high Glycohemoglobin levels together with Family History Diabates and the presence of diabetes (Support: 0.05, Confidence: 0.81, Lift: 6.45). This with other rules provides actionable insights for public health interventions and is especially found to be an important feature affecting the classification outcome.

*3) **Classification Modeling**:* Based on the NHANES dataset, the classification models predicted diabetes risk levels (Normal, Prediabetes, and Diabetes) with varying degrees of accuracy. Table II and III in the Appendix summarizes the performance metrics for each model.

The **Random Forest** was the top-performing model without SMOTE, achieving a test accuracy of 86.84%, supporting H2. This model showed strong performance metrics, particularly in accurately identifying the 'Normal' (F1 Score: 0.93) and 'Diabetes' (F1 Score: 0.71) classes. Support Vector Machine (SVM) and Logistic Regression achieved approximately 86% accuracy, with SVM slightly outperforming Logistic Regression in F1 Scores for the 'Diabetes' class.

All models struggled with the 'Prediabetes' category due to class imbalance, resulting in low F1 Scores (0.00) despite high overall accuracy. **SMOTE** was applied to address this imbalance. In Logistic Regression and SVM, SMOTE marginally improved the F1 Scores for 'Prediabetes' but led to a reduction in overall accuracy from approximately 86% to 73-78%. In contrast, Random Forest with SMOTE maintained a higher accuracy of 83.44% while slightly improving the F1 Score for 'Prediabetes', partially supporting H1, indicating a more balanced performance across classes.

The **Artificial Neural Network** model achieved 86.30% test accuracy without SMOTE, comparable to Random Forest and SVM. However, applying SMOTE reduced accuracy to 68.03%, with a slight improvement in the F1 Score for 'Prediabetes'. This suggests that SMOTE may have introduced noise leading to overfitting or that the neural network required further tuning to handle the synthetic data effectively.

Visualizations, such as confusion matrices and ROC curves, supported these results. The confusion matrices demonstrated the models' ability to distinguish 'Normal' and 'Diabetes' statuses, with high true positive rates for these classes. However, they also highlighted ongoing challenges in accurately classifying 'Prediabetes', with the models producing a high number of false negatives. The ROC curves demonstrated the models' ability to discriminate between classes, with **Random Forest** achieving the highest Area Under the Curve (AUC) for both 'Normal' and 'Diabetes' categories. However, 'Prediabetes' maintained a lower AUC across all models, reflecting the underlying difficulty in prediction due to limited representation in the dataset.

*4) **Feature Importance Analysis**:* Feature importance analysis showed that **Glycohemoglobin (HbA1c)**, **Age**, and **BMI** were the most influential predictors across models, which aligned with clinical understandings of diabetes risk factors. Figure 4 illustrates the top 10 feature importances from the Random Forest model.

Higher HbA1c levels increased the likelihood of being classified as 'Diabetes' in Logistic Regression, whereas Random Forest consistently ranked HbA1c as the top feature, emphasizing its importance as an indicator for long-term blood glucose control and diabetes diagnosis. Furthermore, older age was linked to a higher propensity for diabetes classification in all models, indicating the long-term effects of lifestyle and metabolic changes. Elevated BMI levels were strongly associated with an increased risk of diabetes, supporting the well-documented link between obesity and insulin resistance [14].

Additional features, including family history of diabetes, systolic blood pressure, and diastolic blood pressure, were also found to be significant predictors. In **Random Forest**, **Family History of Diabetes** had a significant impact, indicating the importance of genetic predisposition in diabetes risk. Blood pressure metrics were significant in both **Logistic Regression** and **Random Forest**, indicating the link between cardiovascular health and diabetes.

Dietary factors like calories, total sugars, and fiber were moderately important across models. While not as influential as HbA1c, age, or BMI, these characteristics help to determine the overall risk profile by indicating dietary habits that may affect metabolic health. For example, high sugar consumption is a known risk factor for insulin resistance, whereas adequate fiber intake is associated with better glycemic control [15]

[16]. However, as previously mentioned in the paper [7], it was found to be not significant, and this study confirms that it is not one of the main variables affecting the outcome of the classification.

The consistency of feature importance across models demonstrates the predictors' robustness in assessing diabetes risk. The alignment with clinical knowledge validates the models while also ensuring that the findings are actionable and relevant for healthcare interventions.

V. Discussion

The results show that the Random Forest model outperforms other algorithms in predicting diabetes risk, supporting Hypothesis 2 (H2). The ensemble nature of Random Forests enables them to capture complex, non-linear interactions between features while also reducing overfitting by aggregating multiple decision trees. Random Forests' robustness and ability to handle high-dimensional data make them particularly effective in medical classification tasks, aligning with previous studies that highlight the superiority of ensemble methods in similar contexts [5], [9].

The use of **SMOTE** improved the detection of the 'Prediabetes' class but reduced overall accuracy, partially supporting Hypothesis 1 (H1). While SMOTE improved the F1-Score for 'Prediabetes' across all models, it also reduced overall model accuracy slightly, which was evident in the Logistic Regression and Neural Network models. This finding highlights the ongoing challenge of class imbalance in medical datasets, where minority groups are underrepresented but clinically significant [11].

Our findings align with previous research, indicating that key predictors like **Glycohemoglobin (HbA1c)**, **Age**, and **BMI** are consistently significant. These variables have been repeatedly identified as critical factors in diabetes prediction models [5], [8]. Our study builds on previous research by incorporating complete feature selection, including variables like **Family History of Diabetes** and dietary factors. This enhances the model's predictive capabilities and provides a broader view of risk.

Neural Networks, despite their capacity for modeling complex patterns, did not outperform ensemble methods in this context. The significant drop in accuracy upon applying SMOTE suggests potential overfitting or inadequate representation of minority classes within the network's architecture. Additionally, the interpretability of feature contributions remains limited compared to Random Forest, which readily provides feature importance metrics.

Despite its strengths, this study has several limitations. The NHANES dataset, while comprehensive, lacks certain critical variables that may influence diabetes risk. The Fasting Plasma Glucose Test was missing from the Kaggle dataset, despite being described on the NHAENES website. Other limitations include the absence of key variables such as physical activity levels, dietary quality beyond macronutrients, and psychosocial factors like stress and mental health. The absence of these variables may limit the model's ability to capture the entire range of diabetes risk factors.

Moreover, future work should explore advanced imbalance handling techniques, such as **Balanced Random Forest** or **EasyEnsemble**, which combine undersampling of the majority class with ensemble learning to improve minority class prediction [17]. Additionally, incorporating **cost-sensitive learning** approaches, which assign higher misclassification costs to minority classes, can prioritize the correct classification of 'Prediabetes' and 'Diabetes' cases [18].

Lastly, the practical implications of our findings are important for healthcare providers and policymakers. Integrating the Random Forest model into clinical practice can help identify high-risk individuals for diabetes, allowing for timely interventions and personalized care plans. Furthermore, identifying key risk factors facilitates targeted public health strategies aimed at reducing diabetes prevalence through informed resource allocation and preventive measures.

VI. Conclusion

This study successfully developed and evaluated multiple classification models to predict diabetes risk levels using the NHANES dataset. **Random Forest** emerged as the most effective model, achieving a test accuracy of approximately 86.84% and balanced performance across primary classes. The identification of key predictors—**Glycohemoglobin (HbA1c)**, **Age**, and **BMI**—aligns with established clinical insights, reinforcing the model's validity and applicability in real-world healthcare settings.

The use of **SMOTE** revealed the complexities of addressing class imbalance. Oversampling can improve minority class detection, but it may also introduce trade-offs in overall model accuracy. This emphasizes the importance of sophisticated methods to deal with imbalanced datasets, which may involve combining multiple strategies to optimize performance across all classes.

Clustering and association rule mining improved the analysis by identifying unique patient profiles and highlighting the relationship between inheritance and biological indicators in diabetes risk. These findings not only improve the interpretability of predictive models but also provide actionable information for targeted health interventions and policymaking.

Integrating machine learning models into clinical practice has significant implications for early diabetes detection. Accurate risk prediction allows healthcare providers to implement timely interventions, personalize treatment plans, and allocate resources more efficiently. By identifying high-risk individuals, these models can help to reduce the incidence of severe

complications associated with diabetes, ultimately improving patient outcomes and reducing the load on healthcare systems.

Despite the promising results, this study recognizes several limitations. Excluding variables like physical activity levels and psychosocial factors may limit the model's ability to capture all diabetes risk factors. Future research should concentrate on expanding the feature set to include a wider range of health and lifestyle indicators, which could provide a more complete risk assessment.

In conclusion, this study adds to the increasing amount of evidence that supports the use of machine learning for diabetes risk prediction. The Random Forest model's effectiveness, combined with the identification of key risk factors, demonstrates the potential for data-driven approaches to improve early detection and inform targeted healthcare interventions. Addressing the identified limitations and following the outlined future directions will be critical for advancing predictive modeling in diabetes care and, ultimately, improving public health outcomes.

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APPENDIX

TABLE I
DESCRIPTION OF SELECTED VARIABLES FOR MODELING

Variable Name	Type	Description
Age	Numeric	Participant's age in years.
Gender	Categorical	Gender of the participant (1 = Male, 2 = Female).
Race_Ethnicity	Categorical	Self-reported race/ethnicity (e.g., Non-Hispanic White, Non-Hispanic Black, etc.).
Education_Level	Categorical	Highest level of education completed (e.g., Some College, High School).
Income_Poverty_Ratio	Numeric	Income-to-poverty ratio, an indicator of socioeconomic status.
Systolic_BP_Read1	Numeric	First reading of systolic blood pressure (mmHg).
Diastolic_BP_Read1	Numeric	First reading of diastolic blood pressure (mmHg).
BMI	Numeric	Body Mass Index, an indicator of body fat based on height and weight.
Creatinine	Numeric	Serum creatinine level, a measure of kidney function.
Glycohemoglobin	Numeric	Hemoglobin A1c level, an indicator of blood sugar control over time.
Total_Cholesterol	Numeric	Total cholesterol level (mg/dL).
Calories	Numeric	Daily calorie intake based on dietary data.
Total_Sugars	Numeric	Total sugar intake (grams).
Fiber	Numeric	Total fiber intake (grams).
Family_History_Diabetes	Categorical	Family history of diabetes (Yes/No).
Diabetes_Status	Categorical	Target variable indicating diabetes status (Normal, Prediabetes, Diabetes).

TABLE II
MODEL PERFORMANCE METRICS WITHOUT SMOTE

Model	Accuracy	F1 Score (Normal)	F1 Score (Prediabetes)	F1 Score (Diabetes)
Logistic Regression	86.39%	0.93	0.00	0.68
Support Vector Machine	86.48%	0.93	0.00	0.69
Random Forest	**86.84%**	0.93	0.00	**0.71**
Artificial Neural Network	86.30%	0.92	0.00	0.67

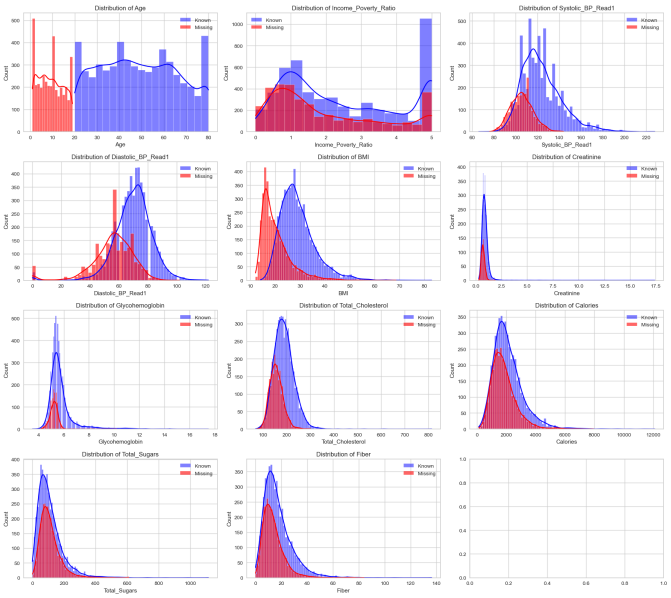


Fig. 2. Distributions of Missing Values for 'Family History Diabetes' Compared to Known Values.

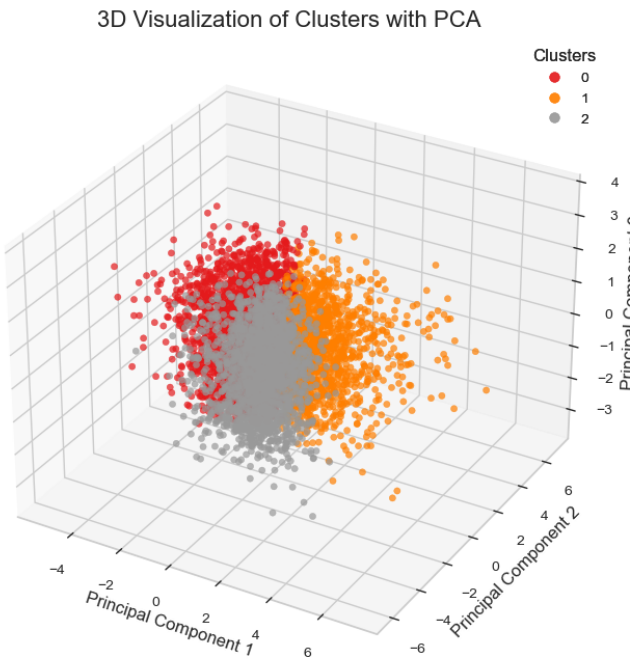


Fig. 3. Clustering Visualization Using PCA.

TABLE III
MODEL PERFORMANCE METRICS WITH SMOTE

Model	Accuracy	F1 Score (Normal)	F1 Score (Prediabetes)	F1 Score (Diabetes)
Logistic Regression	72.69%	0.84	0.18	0.69
Support Vector Machine	78.16%	0.88	0.12	0.63
Random Forest	**83.44%**	0.91	0.09	**0.71**
Artificial Neural Network	68.03%	0.81	0.16	0.67

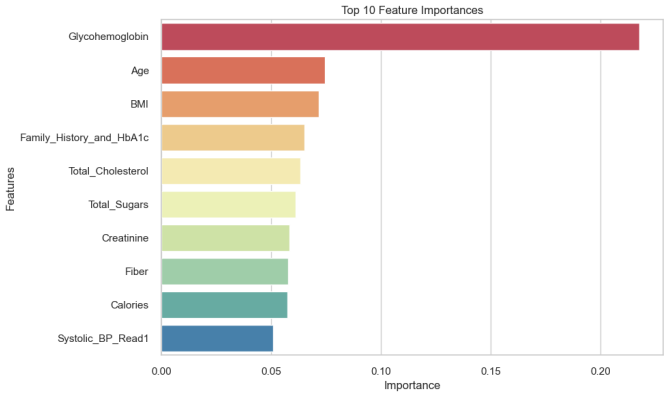


Fig. 4. Top 10 Feature Importances in Random Forest Model